

Checkpoint 3: Baseline U-Net Diffusion Model

Objective

Train and evaluate the baseline conditional U-Net denoising backbone.

Key Message

The baseline model gives the reference point for all later comparisons.

Completed Work

- Implemented conditional U-Net with timestep conditioning.
- Completed baseline training and evaluation on the fixed test split.
- Generated metric CSV files and qualitative comparison grids.

Verification and Controls

- Selected checkpoints from validation behavior without tuning against the held-out test metrics.
- Evaluated clipped predictions with shared MSE, MAE, PSNR, SSIM, and Delta E implementations.
- Retained per-image outputs and a fixed qualitative grid for traceability beyond aggregate means.

Evidence

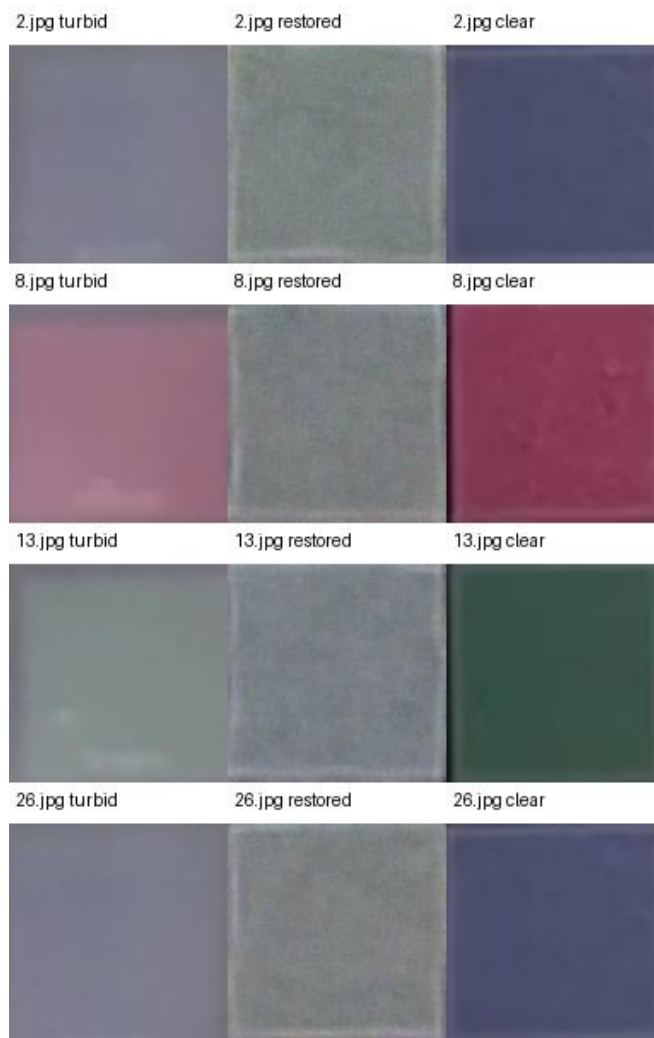
Metric	Baseline mean +/- SD
MSE	0.039873 +/- 0.001756
PSNR	15.780514 +/- 0.187075
SSIM	0.657180 +/- 0.015721
Delta E	28.380208 +/- 1.320879

Decision / Next Gate

Use the default U-Net as the reference architecture and report all later gains relative to this frozen baseline.

Notes

- This model is the default U-Net diffusion baseline, not the proposed contribution.



Baseline restoration examples from the fixed test split.